Model	Acc	Consistency	Consistent-Acc
majority	23.1+-21.0	100.0+-0.0	23.1+-21.0
RoBERTa-med-small-1M	11.2+-9.4	37.1+-11.0	2.8+-4.0
RoBERTa-base-10M	17.3+-15.8	29.8+-12.7	3.2+-5.1
RoBERTa-base-100M	22.1+-17.1	31.5+-13.0	3.7+-5.3
RoBERTa-base-1B	**38.0**+-23.4	**50.6**+-19.8	**18.0**+-16.0

Table 4: Knowledge and consistency results for different RoBERTas, trained on increasing amounts of data. Best model for each measure in bold.

Effect of Pretraining Corpus Size Next, we study the question of whether the number of tokens used during pretraining contributes to consistency. We use the pretrained RoBERTa models from Warstadt et al. (2020) and repeat the experiments on four additional models. These are RoBERTa-based models, trained on a sample of Wikipedia and the book corpus, with varying training sizes and parameters. We use one of the three published models for each configuration and report the average accuracy over the relations for each model in Table 4. Overall, *Accuracy* and *Consistent-Acc* improve with more training data. However, there is an interesting outlier to this trend: The model that was trained on one million tokens is more consistent than the models trained on ten and one-hundred million tokens. A potentially crucial difference is that this model has many fewer parameters than the rest (to avoid overfitting). It is nonetheless interesting that a model that is trained on significantly less data can achieve better consistency. On the other hand, it's accuracy scores are lower, arguably due to the model being exposed to less factual knowledge during pretraining.

6.3 Do PLMs Generalize Over Syntactic Configurations?

Many papers have found neural models (especially PLMs) to naturally encode syntax (Linzen et al., 2016; Belinkov et al., 2017; Marvin and Linzen, 2018; Belinkov and Glass, 2019; Goldberg, 2019; Hewitt and Manning, 2019). Does this mean that PLMs have successfully abstracted knowledge and can comprehend and produce it regardless of syntactic variation? We consider two scenarios. (1) Two patterns differ only in syntax. (2) Both syntax and lexical choice are the same. As a proxy, we define syntactic equivalence when the dependency path between the subject and object are identical. We parse all patterns from PARAREL🤘 using a dependency parser (Honnibal et al., 2020)[12] and retain the path between the entities. Success on (1) indicates that the model's knowledge processing is robust to syntactic variation. Success on (2) indicates that the model's knowledge processing is robust to variation in word order and tense.

Model	Diff-Syntax	No-Change
majority	100.0+-0.0	100.0+-0.0
BERT-base	67.9+-30.3	76.3+-22.6
BERT-large	67.5+-30.2	78.7+-14.7
BERT-large-wwm	63.0+-31.7	**81.1**+-9.7
RoBERTa-base	66.9+-10.1	80.7+-5.2
RoBERTa-large	**69.7**+-19.2	80.3+-6.8
ALBERT-base	62.3+-22.8	72.6+-11.5
ALBERT-xxlarge	51.7+-26.0	67.3+-17.1

Table 5: Consistency and standard deviation when only syntax differs (*Diff-Syntax*) and when syntax and lexical choice are identical (*No-Change*). Best model for each metric is highlighted in bold.

Table 5 reports results. While these and the main results on the entire dataset are not comparable as the pattern subsets are different, they are higher than the general results: 67.5% for BERT-large when only the syntax differs and 78.7% when the syntax is identical. This demonstrates that while PLMs have impressive syntactic abilities, they struggle to extract factual knowledge in the face of tense, word-order, and syntactic variation.

McCoy et al. (2019) show that supervised models trained on MNLI (Williams et al., 2018), an NLI dataset (Bowman et al., 2015), use superficial syntactic heuristics rather than more generalizable properties of the data. Our results indicate that PLMs have problems along the same lines: they are not robust to surface variation.

7 Analysis

7.1 Qualitative Analysis

To better understand the factors affecting consistent predictions, we inspect the predictions of BERT-large on the patterns shown in Table 6. We highlight several cases: The predictions in Example #1 are inconsistent, and correct for the first pattern (*Amsterdam*), but not for the other two (*Madagascar* and *Luxembourg*). The predictions in Example #2 also show a single pattern that predicted the right object; however, the two other patterns, which are

[12] https://spacy.io/

#	Subject	Object	Pattern #1	Pattern #2	Pattern #3	Pred #1	Pred #2	Pred #3
1	Adriaan Pauw	Amsterdam	[X] was born in [Y].	[X] is native to [Y].	[X] is a [Y]-born person.	Amsterdam	Madagascar	Luxembourg
2	Nissan Livina Geniss	Nissan	[X] is produced by [Y].	[X] is created by [Y].	[X], created by [Y].	Nissan	Renault	Renault
3	Albania	Serbia	[X] shares border with [Y].	[Y] borders with [X].	[Y] shares the border with [X]	Greece	Turkey	Kosovo
4	iCloud	Apple	[X] is developed by [Y].	[X], created by [Y].	[X] was created by [Y]	Microsoft	Google	Sony
5	Yahoo! Messenger	Yahoo	[X], a product created by [Y]	[X], a product developed by [Y]	[Y], that developed [X]	Microsoft	Microsoft	Microsoft
6	Wales	Cardiff	The capital of [X] is [Y] .	[X]'s capital, [Y].	[X]'s capital city, [Y].	Cardiff	Cardiff	Cardiff

Table 6: Predictions of BERT-large-cased. "Subject" and "Object" are from T-REx (Elsahar et al., 2018). "Pattern #i" / "Pred #i": three different patterns from our resource and their predictions. The predictions are colored in blue if the model predicted correctly (out of the candidate list), and in red otherwise. If there is more than a single erroneous prediction, it is colored by a different red.

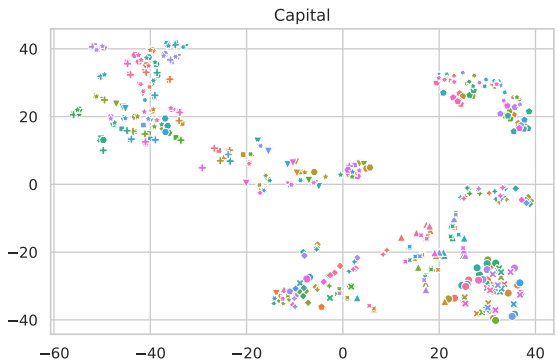

Figure 3: t-SNE of the encoded patterns from the *capital* relation. The colors represent the different subjects, while the shapes represent patterns. A knowledge-focused representation should cluster based on identical subjects (color), but instead the clustering is according to identical patterns (shape).

lexically similar, predicted the same, wrong answer – *Renault*. Next, the patterns of Example #3 produced two factually correct answers out of three (*Greece*, *Kosovo*), but simply do not correspond to the gold object in T-REx (*Albania*), since this is an M-N relation. Note that this relation is not part of the consistency evaluation, but the determinism evaluation. The three different predictions in example #4 are all incorrect. Finally, the two last predictions demonstrate consistent predictions: Example #5 is consistent but factually incorrect (even though the correct answer is a substring of the subject), and finally, Example #6 is consistent and factual.

7.2 Representation Analysis

To provide insights on the models' representations, we inspect these after encoding the patterns.

Motivated by previous work that found that words with the same syntactic structure cluster together (Chi et al., 2020; Ravfogel et al., 2020) we perform a similar experiment to test if this behavior replicates with respect to knowledge: We encode the patterns, after filling the placeholders with subjects and masked tokens and inspect the last layer representations in the masked token position. When plotting the results using t-SNE (Maaten and Hinton, 2008) we mainly observe clustering based on the patterns, which suggests that encoding of knowledge of the entity is not the main component of the representations. Figure 3 demonstrates this for BERT-large encodings of the *capital* relation, which is highly consistent.[13] To provide a more quantitative assessment of this phenomenon, we also cluster the representations and set the number of centroids based on:[14] (1) the number of patterns in each relation, which aims to capture pattern-based clusters, and (2) the number of subjects in each relation, which aims to capture entity-based clusters. This would allow for a perfect clustering, in the case of perfect alignment between the representation and the inspected property. We measure the purity of these clusters using V-measure and observe that the clusters are mostly grouped by the patterns, rather than the subjects. Finally, we compute the spearman correlation between the consistency scores and the V-measure of the representations. However, the correlation between these variables is close to zero,[15] therefore not explaining the models' behavior. We repeated these experiments while inspecting the objects instead of the subjects, and found similar trends. This finding is interesting since it means that (1) these representations are not knowledge-focused, i.e., their main component does not relate to knowledge, and (2) the representation by its entirety does not explain the behavior of the model, and thus only a subset of the representation does. This finding is consistent

[13] While some patterns are clustered based on the subjects (upper-left part), most of them are clustered based on patterns.

[14] Using the KMeans algorithm.

[15] Except for BERT-large whole-word-masking, where the correlation is 39.5 ($p < 0.05$).

with previous work that observed similar trends for linguistic tasks (Elazar et al., 2021). We hypothesize that this disparity between the representation and the behavior of the model may be explained by a situation where the distance between representations largely does not reflect the distance between predictions, but rather other, behaviorally irrelevant factors of a sentence.

8 Improving Consistency in PLMs

In the previous sections, we showed PLMs are generally not consistent in their predictions, and previous works have noticed the lack of this property in a variety of downstream tasks. An ideal model would exhibit the consistency property after pretraining, and would then be able to transfer it to different downstream tasks. We therefore ask: Can we enhance current PLMs and make them more consistent?

8.1 Consistency Improved PLMs

We propose to improve the consistency of PLMs by continuing the pretraining step with a novel consistency loss. We make use of the T-REx tuples and the paraphrases from PARAREL🤘.

For each relation r_i, we have a set of paraphrased patterns P_i describing that relation. We use a PLM to encode all patterns in P_i, after populating a subject that corresponds to the relation r_i and a mask token. We expect the model to make the same prediction for the masked token for all patterns.

Consistency Loss Function As we evaluate the model using acc@1, the straight-forward consistency loss would require these predictions to be identical:

$$\min_{\theta} \text{sim}(\arg\max_{i} f_\theta(P_n)[i], \arg\max_{j} f_\theta(P_m)[j])$$

where $f_\theta(P_n)$ is the output of an encoding function (e.g., BERT) parameterized by θ (a vector) over input P_n, and $f_\theta(P_n)[i]$ is the score of the ith vocabulary item of the model.

However, this objective contains a comparison between the output of two argmax operations, making it discrete and discontinuous, and hard to optimize in a gradient-based framework. We instead relax the objective, and require that the predicted *distributions* $Q_n = \text{softmax}(f_\theta(P_n))$, rather than the top-1 prediction, be identical to each other. We use two-sided KL Divergence to measure similarity between distributions: $D_{KL}(Q_n^{r_i}||Q_m^{r_i}) + D_{KL}(Q_m^{r_i}||Q_n^{r_i})$ where $Q_n^{r_i}$ is the predicted distribution for pattern P_n of relation r_i.

As most of the vocabulary is not relevant for the predictions, we filter it down to the k tokens from the candidate set of each relation (§3.2). We want to maintain the original capabilities of the model – focusing on the candidate set helps to achieve this goal since most of the vocabulary is not affected by our new loss.

To encourage a more general solution, we make use of all the paraphrases together, and enforce all predictions to be as close as possible. Thus, the consistency loss for all pattern pairs for a particular relation r^i is:

$$\mathcal{L}_c = \sum_{n=1}^{k} \sum_{m=n+1}^{k} D_{KL}(Q_n^{r_i}||Q_m^{r_i}) + D_{KL}(Q_m^{r_i}||Q_n^{r_i})$$

MLM Loss Since the consistency loss is different from the Cross-Entropy loss the PLM is trained on, we find it important to continue the MLM loss on text data, similar to previous work (Geva et al., 2020).

We consider two alternatives for continuing the pretraining objective: (1) MLM on Wikipedia and (2) MLM on the patterns of the relations used for the consistency loss. We found that the latter works better. We denote this loss by $\mathcal{L}_{MLM}$.

Consistency Guided MLM Continual Training Combining our novel consistency loss with the regular MLM loss, we continue the PLM training by combining the two losses. The combination of the two losses is determined by a hyperparameter λ, resulting in the following final loss function:

$$\mathcal{L} = \lambda \mathcal{L}_c + \mathcal{L}_{MLM}$$

This loss is computed per relation, for one KB tuple. We have many of these instances, which we require to behave similarly. Therefore, we batch together $l = 8$ tuples from the same relation and apply the consistency loss function to all of them.

8.2 Setup

Since we evaluate our method on unseen relations, we also split train and test by relation type (e.g., location-based relations, which are very common in T-REx). Moreover, our method is aimed to be simple, effective, and to require only minimal supervision. Therefore, we opt to use only three relations: *original-language*, *named-after*, and *original-network*; these were chosen randomly,